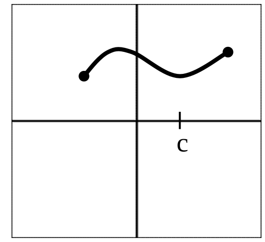
Continuity

Continuous Function

- A function is continuous if it is continuous at each point of its domain.



Continuity at a Point

- A function y = f(x) is continuous at a point, c, if $\lim_{x\to c} f(x) = f(c)$

<u>The Continuity Test:</u>

A function y = f(x) is continuous at x = c if and only if it meets *all* three of the following conditions:

1) f(x) exists (f is defined at c)

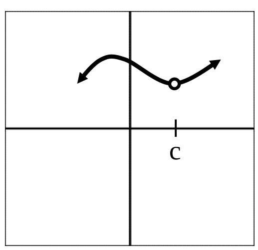
2) $\lim_{x\to c} f(x)$ exists (f has a limit as $x \to c$)

3) $\lim_{x\to c} f(x) = f(c)$ (the limit equals the function value)

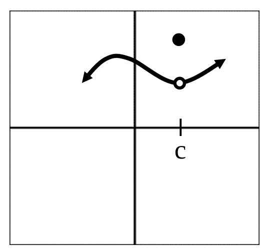
Discontinuous Functions

- If one (or more) of the three conditions is not satisfied, we say that f is discontinuous at c or that f has a discontinuity at c.

a) Point Discontinuity

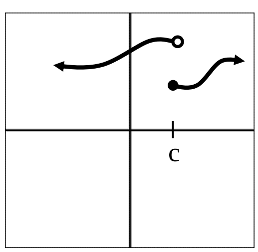


<u>WHY?</u>



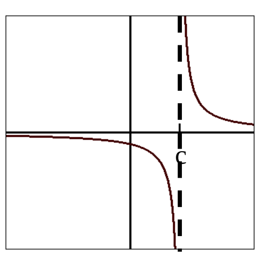
<u>WHY?</u>

b) Jump Discontinuity

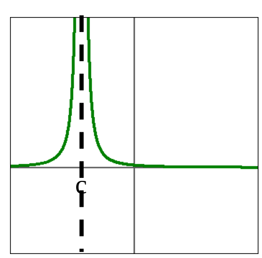


<u>WHY?</u>

c) Infinite Discontinuity



<u>WHY?</u>

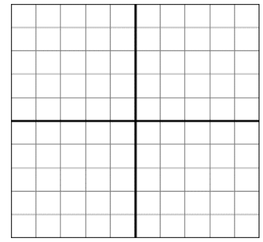


<u>WHY?</u>

A) Removable Discontinuity

- occurs when $\lim\limits_{x \to c} f(x)$ exists but $\lim\limits_{x \to c} f(x) \neq f(c)$
- if you can remove each discontinuity by defining the function value (f(c))

<u>Example:</u> $f(x) = \dfrac{x^2 + x - 6}{x^2 - 4}$



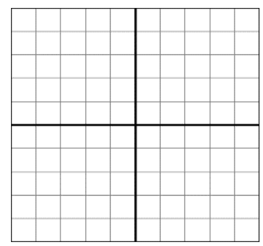
Not defined at x = 2 (or x = -2).

Is x = 2 a removable discontinuity?

- If it is we should be able to find a value to make it continuous at x = 2 (a continuous extension)

$$\lim_{x \to 2} \frac{x^2 + x - 6}{x^2 - 4} =$$

Extended function:

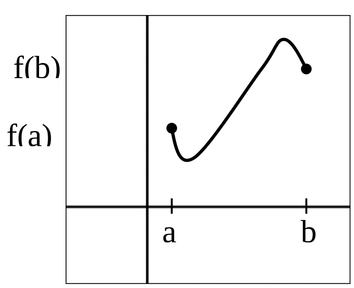


B) Non-removable

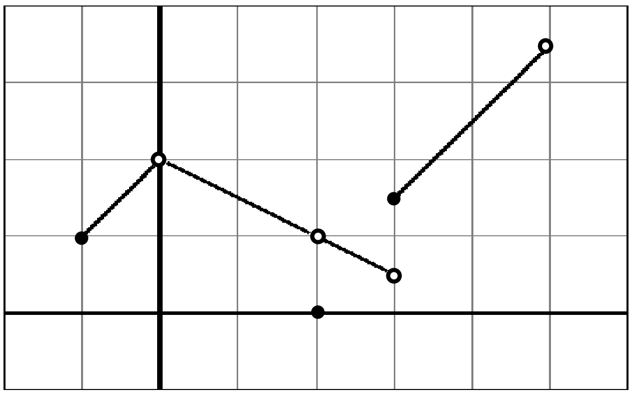
- Occurs when $\lim\limits_{x \to c} f(x)$ does not exist!
- Left and right hand limits do not have the same value.

<u>Intermediate Value Theorem</u>

A function y = f(x) that is continuous on the closed interval [a,b] takes on every value between f(a) and f(b)



Given the following graph, state for what values of x the function is discontinuous and state why it is discontinuous at that point. Also state what type of discontinuity it is and whether it is removable or nonremovable. Explain how any removable discontinuities should be defined or redefined to make the function continuous.



Where?	Why?	Type	Removable (R) or Nonremovable (NR)	If R, what values makes it continuous?

Find the value of a and *k* that makes the function continuous.

$$f(x) = \begin{cases} \dfrac{x^2 - 16}{x+4} & \text{if } x \neq a \\ k & \text{if } x = a \end{cases}$$

a = ______ k = ______

At what points, if any, is the function $f(x) = \dfrac{x}{2x^2 - x - 1}$ discontinuous?